

# Joshua Stuart Research Assignment 1

1. A database is a virtual repository where data is stored, essential for collating & storing related data/information which can later be analysed to support decision making. Examples include a gym franchise e.g. Virgin Active or hospitals e.g. the Netcare franchise
2. A data warehouse is a central database, that stores information/data from many different databases. It differs from a normal db because it stores data from many different db's, a use case is a franchise with a local db for each store, and a data warehouse in head office to store all info from all the local db's
3. A data lake is also a central repository, but this stores unstructured & semi structured data whereas a data warehouse stores structured data. A business would prefer a data lake to store raw data e.g. videos, images, audios, emails, JSON files
4. A data mart is a segment/subset of data (a fraction of a whole dataset) useful for a department. It is a fraction of a data warehouse, typically used by a specific team.
5. Structured data exists in a fixed format, semi structured data has some structure but no fixed format, and unstructured data has no structure or format.
  - \* **Structured Data** - examples include excel spread sheets, great for ensuring data validity
  - \* **Semi Structured Data** - examples include JSON & xml files, lower data validity
  - \* **Unstructured Data** - Audio files, images, videos.

Today's mood:

☐ Craving coffee☐ 11/10☐ Logging off☐ Main character energy☐ BRB overthinking☐ \_\_\_\_\_

To unpack tomorrow:

TYPO

6 Data modelling is a process of visualising the plan of how data should be collected, stored, organized & connected to other data. This visualisation outlines the different types of data as well. Its importance ranges from organization, data accuracy & validity is increased, lower redundancies & easier access & analysis. The 2 types include but are not limited to:

\* **Relational** - Data is organized into a table w rows & columns, connected using primary & foreign keys.

\* **Hierarchical data model** - Data organized in a tree like structure & flows into a one-to-many relationship

7. A process, completed with statistical techniques & machine learning. The objective is to discover meaningful patterns, insights, & relationships from large amounts of data. Businesses use it to make predictions & understand customers & their behaviour.

\* **e.g 1** PnP uses analyses to deduce that customers who buy domestos & sunlight washing liquid are also likely to buy sponges & steelwool, & can recommend those products on online shopping platforms

\* **e.g 2** A bank analysis & tracking a customer's normal expenditure picks up on an anomaly & flags it for fraud.

8. A DBMS (database management system) is a platform that allows you to interact with a data warehouse and pull relevant data from it.

\* **MongoDB** A DBMS platform, stores semi-structured data, NoSQL DB. Great for social media management. Focus in web dev & systems dev

\* **Oracle** A relational DBMS, great for enterprise level operations, OLTP & OLAP. Is a multimodel DBMS

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☐

To unpack tomorrow:

- \* **PostgreSQL** is a relational DBMS famous for complex queries, & strong data integrity. Great for financial systems
- \* **MySQL** Relational DBMS stores data in tables & rows, uses SQL. Great for online stores.

9. ETL is a part of data life cycle & is a data progression system.

**E**xtract data from sources

**T**ransforming data means cleaning data, remove duplicates, data validation etc

**L**oad it to final ~~stage~~ storage place

10. \* **CSV's** (comma separated values) stores data in rows & columns as plain text, using a comma as a delimiter (separator) & is useful for platforms like excel, power BI.

\* **JavaScript Object Notation (JSON)** stores data with a label & value system, e.g: "Name": John. Web dev & APIs are the most common use cases

\* **Parquet** refers to a columnar storage style (information is grouped by columns rather than rows) & is common for datasets & data warehouses.

\* **Avro** is a row based format, great for processing complete records in data pipelines

10. \* **Narrow AI** is AI limited to / built for specific tasks e.g. smart fridge that recognises its contents & makes suggestions

\* **General AI** is strong enough to handle many unrelated tasks

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☐ \_\_\_\_\_

To unpack tomorrow:



## Reference list

1. <https://www.ibm.com/think/topics/data-warehouse-vs-data-lake-vs-data-lakehouse>
2. <https://www.databricks.com/blog/what-is-data-mart>
3. <https://www.ibm.com/think/topics/data-modeling>

Sorry the assignment is incomplete, I literally started a new job on Wednesday and it's a lot of shift work and working weekends as well. No excuses I know but this is what I was able to get done so far 🙄